

Md Takiuddin

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Full-stack engineer with 6+ years shipping production web applications end to end – React/Next.js and TypeScript on the front end; Python (FastAPI/Django), Node.js (NestJS/Express), and Go on the back end. Strong Git, pull-request, and code-review discipline, with a bias toward clean, readable, maintainable code. Comfortable supporting and mentoring junior developers while knowing when to escalate technical decisions, and effective in remote, globally distributed teams.

PROFESSIONAL EXPERIENCE

COGNITUS, AN IBM COMPANY

Senior Software Engineer

Remote

November 2023–Present

- Core full-stack contributor on LambdaX – an enterprise contract-lifecycle-management platform used by large US customers, delivered by a globally distributed remote team
- Designed and shipped agreement-header, line-item, dataset, and modification microservices using Python (FastAPI), SQLAlchemy, and PostgreSQL, exposing clean REST APIs consumed by the React front end and third-party integrations
- Extended the in-house low-code flow orchestrator with reusable action blocks spanning SAP, Outlook, DocuSign, and Teams
- Built a Redis Streams + Kafka event fabric and a Socket.io-to-Elasticsearch audit pipeline aligned with SOC-2 retention
- Mentor junior engineers across the stack through code review and pairing, upholding pull-request and testing standards while escalating cross-cutting technical decisions
- Built an MCP server with FastMCP and an in-app chatbot using PydanticAI with pgvector-backed retrieval

GRYPE DIGITAL

Full-Stack Software Developer

Remote

May 2022–November 2023

- Led a 4-developer full-stack squad for Member Lounge, a SaaS engagement platform built on NestJS, TypeScript, MongoDB, and RabbitMQ with a React/Next.js front end
- Solo-built the HPSA portal full-stack on FastAPI and Next.js, working directly with the CEO on specs, sprint plans, and demos
- Architected tiered subscription billing on Stripe with proration, coupons, webhook processing, and automated invoicing
- Shipped a threaded discussion forum and a queue-driven notification engine across browser push, email, and SMS
- Rolled out MFA (email OTP + trusted-device cookies) and zero-downtime blue/green deploys on Docker Swarm

BUGFIXERS

Full-Stack Developer

Dhaka, Bangladesh

March 2020–April 2022

- Delivered full-stack products across Node.js (Express), Python, React, Flutter, and Electron for clients including LearnGenie, Devsfit, and the Bangladesh Navy
- Built secure offline-capable intranet desktop apps and real-time Socket.io dashboards for operational awareness
- Introduced RabbitMQ + Redis-backed microservice decomposition, GitHub Actions CI/CD, and hardened Linux VPS deployments
- Acted as first responder for production incidents and refactored legacy PHP/Vanilla-JS into modern Express + React stacks

KEY PROJECTS

PROCHESTA – AI EXAM-PREP PLATFORM

CTO & Engineering Lead

Independent Project

2025–Present

- Bangladesh's leading job-exam prep platform (BCS, bank, government) with model tests and performance analytics
- Own the architecture and lead the team through code review across a Next.js web app, Node.js/FastAPI services, and offline-ready Flutter apps
- Built an AI tutor and Bangla AI study assistant with pgvector-backed retrieval
- Own CI/CD, observability, and performance across Node.js, MongoDB, and FastAPI services

TAXSTAR – UAE CORPORATE-TAX SAAS

Solo Architect & Developer

Independent Project

August 2023–February 2024

- Built end-to-end FinTech SaaS helping UAE businesses comply with federal corporate-tax rules
- Implemented rules-driven engine for automated tax calculations, forecasting, and deadline reminders
- Integrated with Xero and Zoho ERP systems to eliminate manual CSV uploads
- Designed multi-tenant architecture ready for horizontal scaling and future VAT/payroll-tax modules

HONEST ELITE – SOCIAL-MEDIA PLATFORM

Full-Stack Developer

Independent Project

September 2022–March 2024

- Developed rich-media social network with real-time chat, voice/video calls, and personalized recommendations
- Built GraphQL API on NestJS microservices with a Next.js SSR front end and WebRTC for calls
- Implemented a collaborative and content-based recommender system to improve feed relevancy and engagement
- Used Redis Streams for real-time messaging and call notifications

TECHNICAL SKILLS

- **Languages:** TypeScript/JavaScript, Python, Go, Rust, SQL, Bash
- **Frontend:** React, Next.js, Tailwind CSS, Flutter
- **Backend & APIs:** FastAPI, NestJS, Express, Django, Flask, gRPC, GraphQL, REST
- **Datastores:** PostgreSQL, MongoDB, Redis, MySQL, Elasticsearch, pgvector
- **Messaging & Streaming:** Apache Kafka, RabbitMQ, Redis Streams, Socket.io
- **Cloud & DevOps:** Docker, Kubernetes, AWS, Cloudflare, NGINX, GitHub Actions, Linux
- **Workflow:** Git, pull-request & code-review workflows, testing (PyTest, Jest), debugging
- **Security & Observability:** Keycloak, OAuth 2.0/OIDC, JWT, MFA, Prometheus, Grafana, OpenTelemetry
- **AI & Data:** LangChain, LangGraph, PydanticAI, FastMCP, scikit-learn, pandas

EDUCATION

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

Bachelor of Science in Mechanical Engineering

Rajshahi, Bangladesh

2016–2022

- Capstone: "Low-Cost Autonomous Line-Follower Robot for Warehouse Logistics" – 1st runner-up in RUET TechFest 2019
- Technical Secretary, Robotic Society of RUET; software lead for Team Scrapper (Indian Rover Challenge)
- Coursework: Thermodynamics, Fluid Mechanics, CAD/CAM, Control Systems, Materials Science, Industrial Automation

LANGUAGES

- Bangla (Native), English (Full Professional), Hindi (Limited Working)